STATISTICS · STATISTICAL INFERENCE

Significance Test & Power Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Significance Test & Power Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict

before you touch the tool

1. A coin lands heads 60 times in 100 flips. If the coin is fair, how unusual is 60 or more heads? Estimate.
   Fairly unusual: z = 2, so P($\hat{p}$ ≥ 0.60) ≈ 0.02.
2. If H_0 is rejected at $\alpha = 0.05$, must it also be rejected at $\alpha = 0.01$? Explain.
   No. Only if the p-value is also ≤ 0.01.

2 Explore

use the tool

1. H_0: $p = 0.5$, H_a: $p > 0.5$, $n = 100$. Find the smallest number of successes that is significant at $\alpha = 0.05$. Record z and the p-value.
   59 successes: z = 1.8, p-value ≈ 0.0359. (58 gives 0.0548.)
2. With 59 successes, switch to H_a: $p \neq 0.5$. Record the p-value. Is the result still significant?
   ≈ 0.0719 (doubled). Not significant at α = 0.05.
3. Keep $\hat{p} = 0.55$. Record the one-sided p-values for n = 100, 200 and 400.
   0.1587, 0.0786, 0.0228. Only n = 400 is significant.
4. Errors & power: $p_0 = 0.5$, H_a: $p > 0.5$, $\alpha = 0.05$, true $p = 0.6$. Find the smallest n on the slider with power ≥ 0.80.
   n = 160 (power ≈ 0.817; n = 150 gives 0.794).

3 Record

use the tool and fill in the table

Power (H_0: $p = 0.5$, H_a: $p > 0.5$)

Use the Errors & power tab.

n	α	True p	Power	β = P(Type II error)
100	0.05	0.60	0.6415	0.3585
150	0.05	0.60	0.7942	0.2058
100	0.01	0.60	0.3695	0.6305
100	0.05	0.55	0.2585	0.7415
100	0.05	0.50	0.05 (= α)	— (H_0 is true)

4 Explain

what did you discover?

1. The p-value is the probability, assuming **H_0** is true, of a result at least as **extreme** as the one observed.
2. If the p-value ≤ α, we **reject** H_0. Otherwise we **fail to reject** H_0.
3. A Type I error is **rejecting a true H_0**. A Type II error is **failing to reject a false H_0**.

4. Power increases when n **increases**, when α **increases**, or when the true value is **farther from** p_0.

5. A study reports a p-value of 0.30 and concludes, "The null hypothesis is true." What is wrong with this conclusion?

   A large p-value only means the data are not convincing evidence against H_0. We fail to reject H_0; we do not prove it. The study may have too little power to detect a real difference.

5 Apply a new situation, no tool

1. H_0: $p = 0.30$, H_a: $p > 0.30$. An SRS of 200 has 72 successes. Find z and the p-value, and state a conclusion at $\alpha = 0.05$.

   $\hat{p} = 0.36$, SE $= \sqrt{(0.3 \cdot 0.7/200)} \approx 0.0324$, $z \approx 1.85$, p-value $\approx 0.032 < 0.05$: reject H_0; convincing evidence that $p > 0.30$.

2. H_0: a new drug is no better than the old one. Describe a Type I error and a Type II error in context, and a consequence of each.

   Type I: conclude the new drug is better when it isn't (patients switch for no benefit). Type II: miss a drug that really is better (patients lose a better treatment).

3. How does changing α from 0.01 to 0.05 affect the probability of a Type I error and the power?

   P(Type I) rises from 0.01 to 0.05, and power increases (β decreases).

Teacher notes

Intended discovery: The p-value is a tail area under the null model; the same $\hat{p}$ becomes more significant as n grows; power rises with n, α and effect size, and lowering α raises β.

Likely misconceptions:

- The p-value is the probability that H_0 is true.
- Failing to reject H_0 proves H_0.
- Two-sided and one-sided p-values are the same.

Answers that can vary: Explain wording; Apply 2 consequences. All numbers use the normal approximation (stated in the tool).